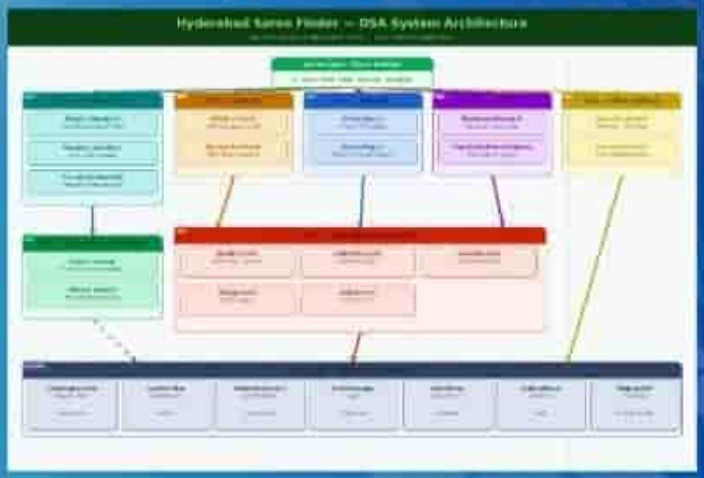
PreSareeFinder – Budget-Based Saree Discovery Platform for Hyderabad

Project Based Learning From DSA

Abstract

The Pre Saree Finder DSA System is a Java-based application developed to demonstrate the practical implementation of Data Structures and Algorithms. The system allows users to search, view, and manage saree information efficiently. Various data structures such as arrays, linked lists, stacks, queues, and hash tables are used to organize and manage saree data. Searching and sorting algorithms help retrieve and arrange saree details quickly. The project highlights how DSA concepts can be applied to build efficient programs for data management and real-world applications.

System Architecture



CO - WISE IMPLEMENTATION

CO1: Algorithm Analysis and Complexity

Analysed problem statements and evaluated algorithm efficiency using Big-O notation and time complexity concepts.

CO2 – Searching & Sorting

Implemented searching and sorting algorithms (Linear Search, Binary Search, Bubble Sort, Selection Sort, Insertion Sort, Merge Sort, Quick Sort) to manage saree data efficiently.

CO3 – Arrays / Linked Lists

Designed and implemented Abstract Data Types using arrays and linked lists to store and manipulate saree information.

CO4 – Stack / Queue

Applied stacks and queues for functionalities like wishlist management and order processing.

CO5 – Hashing / Priority Queue

Implemented hash-based data structures and priority queues to enable faster data retrieval and prioritized operations.

CO6 – Complete Application

Developed a real-world Java application using Data Structures and Algorithms to manage saree data, search operations, and cart management efficiently.

Conclusion and Future Scope:

- The project demonstrates the practical use of Data Structures and Algorithms to manage saree data efficiently using searching, sorting, stacks, queues, and other structures, improving data handling and performance.
- Future Scope:
- The system can be improved by connecting it to a database for permanent data storage and expanding it to include saree stores from multiple locations.

Name: Kamnam Priyanka
Roll no: 2520090001
Section: 8
Cluster: 2